

Notes: Fillmore (RES, 2023)
Price Discrimination and Public Policy in the US College Market

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1 Big picture and incremental contribution

- **Question.** What are the welfare and allocation consequences of giving colleges FAFSA information that can be used in individualized pricing?
- **Mechanism.** FAFSA information helps colleges infer willingness to pay and competitive pressure. This enables price discrimination, not just distribution of aid.
- **Reduced-form facts.** Higher parent income predicts higher transaction prices; higher student desirability predicts lower prices; more schools listed on FAFSA predicts lower prices.
- **Structural model.** College pricing is recast as a first-price auction in utility bids.
- **Main trade-off.** Restricting FAFSA information lowers prices for many students who remain in elite colleges, but up to 13% of students are inefficiently priced out of the elite sector.
- **Distributional result.** FAFSA use transfers resources from higher-income to lower-income students on average, but the redistribution is incomplete and imprecise.

1.1 Incremental contribution of the paper

- The paper is the first to quantify the consequences of allowing colleges to use FAFSA information in pricing.
- It differs from college-market matching models by keeping price discrimination central while restricting only specific information sets.
- It separates student willingness to pay, college desirability of the student, and competition among colleges.
- It turns a data-sharing policy question into a market-equilibrium problem with redistribution and misallocation.
- It contributes methodologically by adapting first-price auction identification to a setting where utility bids are latent but transaction prices are observed.

2 Data and measurement

2.1 NPSAS 2007–08 and transaction prices

The main data are the 2007–08 National Postsecondary Student Aid Study. The transaction price is sticker price minus institutional discount. Federal grants are excluded because they are portable across colleges.

Interpretation: The data observe student-level prices and FAFSA variables, which allows the paper to study within-college price discrimination directly.

2.2 Sample construction

The full sample contains traditional college students under age 31, US citizens or residents, in the 50 states plus D.C. Athletes and employment-related tuition waivers are removed. The freshman sample restricts to dependent freshmen. The elite sample restricts to private and very selective public colleges.

Interpretation: The elite sample is the relevant price-discrimination market because Figure 1 and reduced-form evidence show that strategic discounts are concentrated there.

2.3 BPS 2003–04 and applications

BPS provides actual applications, which are not observed in NPSAS. It is used to estimate how the number of schools listed on FAFSA predicts the true number of applications.

Interpretation: The FAFSA list is treated as a noisy signal of competition, not as the true number of bidders.

3 Model and empirical specifications

3.1 Student utility

$$u_{ij} = v_{ij} - p_{ij}.$$

Interpretation: A student chooses the college that provides the highest utility relative to the non-elite outside option.

3.2 Auction in utility bids

A college offers a utility bid $\beta(s | X)$, where s is match surplus. The equilibrium bid solves an ordinary differential equation of the first-price-auction form:

$$\beta'(s | X) = (s - \beta(s | X)) \frac{\sum_n (n-1) F(s | X)^{n-1} f(s | X) \rho(n | X)}{\sum_n F(s | X)^n \rho(n | X)}, \quad \beta(0 | X) = 0.$$

Interpretation: The ODE implies that higher match surplus raises the student utility bid, but colleges keep part of the surplus.

3.3 Reduced-form pricing equation

$$\text{Tuition discount}_{ij} = X_i \beta + a_j + e_{ij}.$$

Interpretation: College fixed effects compare students within the same college and test the model’s qualitative predictions.

3.4 Counterfactuals

The five counterfactuals restrict parent income, number of schools listed, and FAFSA completion in different combinations. Colleges can still use remaining covariates to proxy for missing information.

Interpretation: The policy experiment is about information sets, not about eliminating all price discrimination.

4 Identification and estimation strategy

- The market is restricted to elite colleges because this is where individualized discounting is important.
- Reduced-form estimates validate the structural model’s qualitative predictions.

- BPS estimates $\rho(n | X)$, the distribution of potential bidders conditional on FAFSA-listed schools.
- Quantile regressions estimate the left tail of the price distribution and recover college willingness to receive.
- Auction equilibrium conditions map observed prices into college payoffs, match surplus, and student surplus.
- Counterfactuals forecast restricted FAFSA variables from remaining covariates and simulate new equilibrium prices.
- Main threats include unobserved auction heterogeneity, endogenous FAFSA lists, and ambiguity over how colleges use marginal surplus.

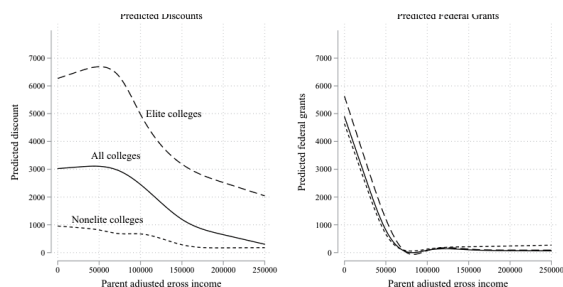
5 Tables and figures

5.1 Figure 1: Relationship between parent AGI and tuition discounts and federal grants + Table 1: Descriptive statistics

Read: Figure 1 shows that elite-college discounts remain positive at high incomes while federal grants vanish around middle incomes. Table 1 shows elite colleges have higher sticker prices and larger discounts.

Interpretation: Institutional discounts are not the same as federal need-based aid; they are strategic prices.

Economic message: FAFSA data let elite colleges price discriminate beyond the federal grant formula.



The fitted values plotted here come from regressing discounts and federal grants on student covariates. Regressions were estimated on the full freshman sample and on elite and nonelite colleges separately. In all six regressions, the fitted values represent a white female student with all other covariates set to their sample means. Parent adjusted gross income is included as a cubic spline with knots at \$25k, \$50k, \$75k, \$100k, \$150k, and \$200k. Sample weights were used in all regressions (NPSAS variable WTA000).

SOURCE: U.S. Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2000

FIGURE 1
Relationship between parent AGI and tuition discounts and federal grants

(a) Relationship between parent AGI and discounts/grants

	Freshmen sample				Elite sample			
	Mean	Std Dev	Min	Max	Mean	Std Dev	Min	Max
Sticker price	\$10,168	\$9,737	\$275	\$39,289	\$20,591	\$9,676	\$1,687	\$39,289
Tuition discount	\$2,914	\$5,722	\$0	\$37,000	\$7,432	\$7,739	\$0	\$37,000
Transaction price	\$7,254	\$7,170	-\$11,771	\$38,678	\$13,159	\$8,811	-\$11,771	\$38,678
Student received positive discount	0.38	0.49	0	1	0.69	0.46	0	1
Tuition discount (if positive)	\$7,635	\$7,052	\$100	\$37,000	\$10,740	\$7,142	\$100	\$37,000
Parent adjusted gross income	\$65,400	\$55,502	\$0	\$496,347	\$78,103	\$64,883	\$0	\$490,000
ACT score	21.2	4.7	11	36	23.5	4.6	11	36
High school GPA	3.42	-0.57	0.90	4.00	3.63	-0.47	1.40	4
Earned AP credit	0.22	0.42	0	1	0.31	0.46	0	1
Parents with college degree	0.68	0.80	0	2	0.91	0.84	0	2
Completed FAFSA	0.82	0.38	0	1	0.85	0.35	0	1
Additional colleges listed on FAFSA	1.2	1.7	0	5	1.9	1.9	0	5
Age as of 31 December 2007	18.6	0.9	17	23	18.4	0.7	17	23
Female	0.54	0.50	0	1	0.56	0.50	0	1
White	0.65	0.48	0	1	0.68	0.47	0	1
Black	0.13	0.34	0	1	0.11	0.31	0	1
Hispanic	0.12	0.33	0	1	0.10	0.31	0	1
Asian	0.05	0.22	0	1	0.07	0.25	0	1
Other	0.04	0.20	0	1	0.04	0.20	0	1

Notes: Sticker prices come from NPSAS variable TUITION2. Tuition discounts come from NPSAS variable INGR2AM. The transaction price is the difference between sticker price and the student's discount. The prices reported apply to all college the student actually attended. Additional Colleges Listed on FAFSA is only calculated for those students who completed the FAFSA. Students can list up to six colleges on the FAFSA, so the number of additional colleges listed runs from 0 to 5. The elite sample is a subset of the freshmen sample and consists of freshmen at private and selective public 4-year colleges. No sample weights were used. Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2000

(b) Descriptive statistics

5.2 Table 2: Within-college discounting and federal aid patterns

Read: Parent income lowers discounts; ACT/GPA and FAFSA-listed schools raise discounts; federal grants are much more need-based.

Interpretation: The same college charges different net prices based on willingness to pay, desirability, and perceived competition.

Economic message: Elite-college institutional aid is a competitive pricing instrument.

TABLE 2
Within-college discounting and federal aid patterns

Dependent variable	Tuition discount		Federal grants
Parent AGI (in \$10,000's)	-123.5 (15.22)***	-88.9 (19.6)***	-380.3 (9.8)***
Number of additional colleges listed on FAFSA	372.7 (56.64)***	366.3 (56.09)***	39.1 (15.30)*
Completed FAFSA	850.0 (164.3)***	870.3 (163.6)***	522.3 (55.43)***
ACT score	133.4 (18.43)***	130.5 (18.53)***	-3.5 (6.062)
High school GPA	394.1 (101.1)***	384.7 (100.8)***	-5.0 (45.29)
Earned AP credit in high school	277.7 (174.2)	292.1 (174.2)	17.2 (51.18)
Number of parents with college degree	93.8 (90.59)	114.6 (91.07)	-67.5 (27.45)*
Age as of 31 December 2007	-37.5 (58.41)	-18.6 (58.25)	-86.9 (30.02)**
Female	113.3 (115.6)	111.1 (115.5)	47.9 (40.93)
Black	580.2 (232.0)*	649.8 (232.8)**	340.6 (93.32)***
Hispanic	581.0 (245.8)*	635.2 (245.7)**	228.0 (86.48)**
Asian	44.1 (391.9)	91.2 (390.2)	278.4 (104.4)**
Other/Multiple	549.7 (360.7)	517.0 (358.1)	100.5 (108.5)
Out-of-state, public	1,135.5 (295.4)***	1,162.4 (295.3)***	61.7 (97.12)
College fixed effects	Yes	Yes	Yes
Parent AGI included as cubic spline	No	Yes	Yes
Observations	5,640	5,640	5,640
R ²	0.670	0.672	0.718

Notes: The regressions include students from the freshmen sample. The omitted race category is "white." For students who completed the FAFSA, "number of additional colleges listed" ranges from 0 to 5 (student can list up to six colleges on the FAFSA). For those who did not complete the FAFSA, "number of additional colleges listed" is set to 0 and the dummy "completed FAFSA" is included. In Columns 2 and 3, Parent Adjusted Gross Income is included as a cubic spline with knots at \$25k, \$50k, \$75k, \$100k, \$150k, and \$200k, and both columns report average partial effects for income (Figure 1 plots the full spline function). Robust standard errors are in parentheses. Sampling weights were used (NPSAS variable WTA000)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid

Within-college discounting and federal aid patterns

5.3 Table 3: Number of colleges listed on FAFSA + Table 4: Colleges listed and month FAFSA completed

Read: FAFSA-listed schools predict discounts and applications; the timing evidence is strongest for early filers.

Interpretation: The FAFSA list is an informative but noisy signal of competition.

Economic message: Colleges respond to perceived competition.

TABLE 3
The number of colleges listed on a student's FAFSA

Dependent variable: tuition discount							
Data set			NPSAS 07-08	NPSAS 07-08	BPS 03-04	BPS 03-04	BPS 03-04
Number of additional colleges listed on FAFSA			372.7 (56.64)***		269.4 (37.36)***	267.3 (39.59)***	
Number listed on FAFSA × Very selective	Public			131.8 (90.02)			246.9 (84.39)**
	Private			909.8 (239.9)***			344.0 (149.4)*
	Moderately selective	Public		56.0 (62.38)			87.9 (49.34)
		Private		249.3 (155.8)			148.5 (104.3)
	Not selective	Public		44.9 (104.6)			−114.3 (87.57)
		Private		264.5 (307.7)			157.0 (196.6)
	Two-year	Public		−0.5 (36.80)			−7.2 (52.15)
		Private		−463.8 (204.5)*			91.8 (411.1)
Number of colleges applied to						3.9 (26.64)	
Number of colleges applied to × Very selective	Public						−31.5 (43.38)
	Private						67.0 (75.13)
	Moderately selective	Public					86.0 (36.85)*
		Private					−53.4 (77.07)
	Not selective	Public					75.5 (71.52)
		Private					23.5 (163.0)
	Two-year	Public					−4.7 (30.39)
		Private					20.0 (280.1)
College fixed effects			Yes	Yes	Yes	Yes	Yes
Observations			5,640	5,640	5,290	5,290	5,260
R^2			0.670	0.714	0.640	0.640	0.678

Notes: See notes to Table 2. The regressions reported in Columns 1 and 2 are identical to the first regression in Table 2 except that in Column 2 the number of additional colleges listed on the FAFSA has been interacted with college selectivity. The remaining covariates were included but not reported here. Column 3 reports estimates from the same regression specification as in column 1, but using data from BPS:2003–4. The reported coefficient in Column 3 is smaller than in Column 1 partly because the dependent variable is expressed in current dollars (no adjustment for inflation). In Column 4, the number of colleges the student actually applied to was included as an additional control (this variable is available in BPS but not in NPSAS). In Column 5, the number of additional colleges listed on the FAFSA and the number of college applications were both interacted with college type. The number of colleges listed on the FAFSA is set to 0 for students who did not complete the FAFSA and a dummy for whether the student completed the FAFSA is included. In Columns 2

(a) Number of colleges listed on FAFSA

TABLE 4
Colleges listed and month FAFSA completed

Dependent variable: tuition discount		
Number of additional colleges listed on FAFSA	January	530.6 (144.2)***
	February	290.7 (81.7)***
	March	11.3 (91.5)
	April or later	77.4 (95.1)
College fixed effects		Yes
Observations		5,630
R^2		0.673

Notes: See notes to Table 2. The regression specification here is identical to Column 1 of Table 2 except that the month the student's FAFSA was received is interacted with the number of colleges listed on her FAFSA. Dummies for month were included but not reported: 760 students applied in January, 1,400 in February, 850 in March, and 1,540 in April or later. 1,080 students did not complete the FAFSA (the omitted category). The remaining covariates from Table 2 were included but not reported here. Robust standard errors are in parentheses. Sampling weights were used (NPSAS variable WTIA000).
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008.

(b) Colleges listed and month FAFSA completed

5.4 Figures 2–3: FAFSA lists and application distributions

Read: The number of schools listed on FAFSA shifts the distribution of applications.

Interpretation: FAFSA lists provide a useful signal about how many bidders a college may face.

Economic message: Removing the FAFSA list removes competition information.

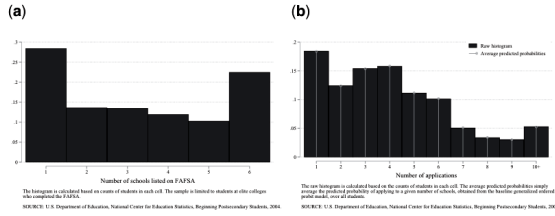


FIGURE 2
Histograms of colleges listed on FAFSA and college applications. (a) Histogram of colleges listed on FAFSA. (b) Histogram of number of applications

(a) Colleges listed and applications

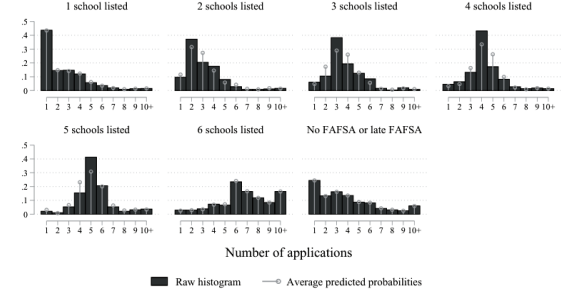


FIGURE 3
Histogram of applications given schools listed on FAFSA

(b) Applications conditional on FAFSA list

5.5 Figure 4: Estimated structural objects + Table 5: Student share of match surplus

Read: Figure 4 shows willingness-to-receive, college payoffs, and willingness-to-pay. Table 5 shows the student share rises from 16.9% with one school listed to 41.9% with six schools listed.

Interpretation: More perceived competition shifts surplus toward students.

Economic message: FAFSA list information affects the surplus split.

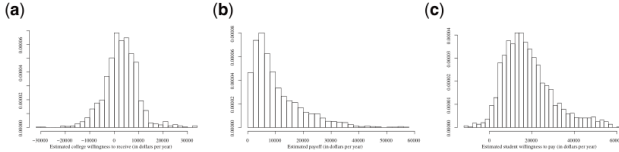


FIGURE 4
Histograms of estimated college willingness to receive, college payoffs, and student willingness to pay
Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004. (a) Willingness to receive ($-w_{ij}$). (b) College payoffs (π_{ij}). (c) Willingness to pay (v_{ij}).

where $\alpha_1(X, j) > 0$, $\alpha_2(X, j) > 0$, and $p(X, j)$ are all parameters to be estimated. This quadratic form

(a) Estimated structural objects

TABLE 5 Student share of match surplus		
	No FAFSA	Average student share of match surplus (%)
Number of schools listed		
on FAFSA	1	16.9
	2	32.6
	3	37.7
	4	40.2
	5	40.4
	6	41.9

Notes: Each cell reports the average student share of the match surplus for students in that cell. Students who complete the FAFSA after March 31 are included in the "No FAFSA" cell. No sample weights were used.
Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008.

(b) Student share of match surplus

5.6 Tables 6–7: Proxies for missing information and predicting applications

Read: Student covariates are imperfect proxies for income and FAFSA-list information. Without FAFSA-list variables, prediction of applications worsens sharply.

Interpretation: The counterfactual information restrictions create real uncertainty for colleges.

Economic message: Regulation has bite because proxies are imperfect.

Counterfactual	Information available to colleges				Dependent variable		
	Core covariates	Parent income	Number of colleges listed	Whether completed FAFSA	Parent income	Number of colleges listed	Whether completed FAFSA
1	Yes	Poverty Dum	Yes	Yes	31.9% (46,472)	100.0% (0)	100.0% (0)
2	Yes	No	Yes	Yes	16.3% (51,508)	100.0% (0)	100.0% (0)
3	Yes	Yes	No	Yes	100.0% (0)	8.0% (1.64)	100.0% (0)
4	Yes	Poverty Dum	No	No	29.9% (46,906)	6.0% (0.370)	6.0% (0.370)
5	Yes	No	No	No	12.2% (52,493)	—	3.7% (0.374)

Notes: This table reports the R^2 from regressing parent income, the number of colleges listed on the FAFSA, and whether student completed the FAFSA on the remaining student covariates. The root mean squared error is reported in parentheses. The table provides a sense for the degree to which other student covariates are able to serve as proxies for the FAFSA variables. In Rows 1 and 4, I assume that colleges, in lieu of using actual income, are instead able to use whether income is above or below the federal poverty line for a family of four. In Rows 2 and 5, I assume that colleges are unable to use any direct measures of income. In Row 3, the regression for number of colleges listed is restricted to students who completed the FAFSA. In Rows 4 and 5, I assume that colleges are unable to use any information from the FAFSA, and since colleges must pool the FAFSA and non-FAFSA students together, in Rows 4 and 5, I do not report an R^2 for the number of colleges listed. No sample weights were used.

Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008.

(a) Proxies for missing information

Counterfactual	Information used in prediction				Goodness of fit	
	Core covariates	Parent income	Number of colleges listed	Whether completed FAFSA	Pseudo R^2 (%)	Root-MSE
1	Yes	Poverty Dum	Yes	Yes	28.9	2.162
2	Yes	No	Yes	Yes	28.8	2.163
3	Yes	Yes	No	Yes	7.2	2.458
4	Yes	Poverty Dum	No	No	5.3	2.479
5	Yes	No	No	No	5.2	2.479
Baseline	Yes	Yes	Yes	Yes	29.5	2.154
	No	No	Yes	Yes	27.2	2.184

Notes: The table reports results from ordered probit regressions of the number of college applications on student covariates using data from BPS:2003–4. Rows 1–5 correspond to the five counterfactuals in the structural estimation. Row 6 represents the baseline prediction. Row 7 represents the prediction using only the number of colleges listed on the FAFSA and FAFSA completion. Comparing rows 6 and 7, we see that including other student covariates modestly improves the prediction of the number of applications. In Rows 1, 2 and 6, 7, a separate ordered probit regression was run for each level of the number of college listed on the FAFSA as well as for students who did not complete the FAFSA. In Row 3, separate regressions were run for students who completed the FAFSA and those that did not. In Rows 4 and 5, all students were pooled in the same regression. The pseudo R^2 is computed by calculating each observation's predicted (expected) number of applications and taking the sum of squared differences between the observed and predicted number of applications. The same quantity is calculated for a "null" model in which no student covariates were used. Finally, the pseudo R^2 is computed as one minus the ratio of these two quantities. The root-MSE is computed by dividing the sum of squared differences by the residual degrees of freedom and taking the square root. Sample weights were not used.

(b) Predicting applications

5.7 Tables 8–9: Information loss and counterfactual estimates

Read: Restricting income and FAFSA-list information raises forecast errors, lowers prices for stayers, raises student surplus, and prices up to 13% of students out of the elite sector.

Interpretation: Lower prices transfer surplus to students, but information loss reduces allocative efficiency.

Economic message: Information restrictions help students on average but create misallocation.

Core covariates	Information available to colleges				Standard deviation of forecast error	
	Parent income	Number of colleges listed	Whether completed FAFSA		Error for forecast of student valuation (e)	Error for forecast of college valuation (z)
Yes	Poverty Dum	Yes	Yes		2.46	0.38
Yes	No	Yes	Yes		2.65	0.42
Yes	Yes	No	Yes		1.55	0.00
Yes	Poverty Dum	No	No		2.94	0.38
Yes	No	No	No		3.33	0.42

Notes: The table reports the standard deviation of the forecast errors in each of the five counterfactuals. Note that, I assume that neither the number of colleges listed on the FAFSA nor whether the student completed the FAFSA directly affect the college's valuation for the student. Thus, for z, the standard deviation in counterfactual 3 is zero and Rows 4 and 5 are identical to Rows 1 and 2. No sample weights were used.

Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.

(a) Information loss

Counterfactual	Panel A: Levels				Baseline
	1	2	3	4	
Consumer surplus (student) surplus per student	\$5,072	\$5,046	\$5,097	\$5,041	\$5,072
Consumer surplus (student) surplus per student	(\$4,105, \$5,666)	(\$4,738, \$5,666)	(\$4,595, \$5,452)	(\$4,458, \$5,716)	(\$4,105, \$5,666)
Total surplus per student	\$15,417	\$15,139	\$15,279	\$15,181	\$15,417
Of those who remain at the college:	(\$13,770, \$15,555)	(\$13,668, \$15,449)	(\$13,664, \$15,428)	(\$13,623, \$15,285)	(\$13,770, \$15,555)
Mean student share of surplus	30.1%	37.5%	37.7%	36.7%	30.1%
Mean student share of surplus	(\$13,066)	(\$13,595, \$8,195)	(\$13,595, \$8,195)	(\$13,095, \$8,645)	(\$13,066)
Mean transaction price	(\$12,818, \$13,330)	(\$12,757, \$13,541)	(\$12,778, \$13,544)	(\$12,980, \$13,790)	(\$12,818, \$13,330)
Panel B: Changes relative to baseline	\$627	\$673	\$481	\$755	\$627
Consumer (student) surplus per student	(\$443, \$733)	(\$485, \$802)	(\$261, \$538)	(\$566, \$981)	(\$443, \$733)
Total surplus per student	(\$0)	(\$161, \$-45)	(\$-215, \$-57)	(\$-174, \$-30)	(\$0)
Percent of students who indirectly choose a non-elite college	0.0%	0.0%	0.0%	0.0%	0.0%
Percent of students who remain at elite colleges:	—	—	—	—	—
Mean change in student share of surplus	6.1%	5.8%	4.2%	5.9%	6.1%
Mean change in transaction price	(\$-15, \$-95)	(\$-15, \$-95)	(\$-15, \$-95)	(\$-15, \$-95)	(\$-15, \$-95)
Percent of students with price drop	0.0%	0.0%	0.0%	0.0%	0.0%
Within-college variance in price (in millions)	38.81	38.81	38.81	38.81	38.81
Number of schools based on FAFSA	Yes	Yes	Yes	Yes	Yes
Parent income	Yes	Yes	Yes	Yes	Yes
FAFSA information available	Yes	Yes	Yes	Yes	Yes
Whether completed FAFSA	Yes	Yes	Yes	Yes	Yes

Notes: The table reports results from ordered probit regressions of the number of college applications on student covariates using data from BPS:2003–4. Rows 1–5 correspond to the five counterfactuals in the structural estimation. Row 6 represents the baseline prediction. Row 7 represents the prediction using only the number of colleges listed on the FAFSA and FAFSA completion. Comparing rows 6 and 7, we see that including other student covariates modestly improves the prediction of the number of applications. In Rows 1, 2 and 6, 7, a separate ordered probit regression was run for each level of the number of college listed on the FAFSA as well as for students who did not complete the FAFSA. In Row 3, separate regressions were run for students who completed the FAFSA and those that did not. In Rows 4 and 5, all students were pooled in the same regression. The pseudo R^2 is computed by calculating each observation's predicted (expected) number of applications and taking the sum of squared differences between the observed and predicted number of applications. The same quantity is calculated for a "null" model in which no student covariates were used. Finally, the pseudo R^2 is computed as one minus the ratio of these two quantities. The root-MSE is computed by dividing the sum of squared differences by the residual degrees of freedom and taking the square root. Sample weights were not used.

(b) Counterfactual estimates

5.8 Table 10 and Figure 5: Counterfactual pricing patterns and income gradient

Read: When income is restricted, the price-income gradient flattens but does not disappear because other covariates proxy for income.

Interpretation: Direct data bans do not eliminate price discrimination when correlated proxies remain.

Economic message: Policy must consider substitution into alternative signals.

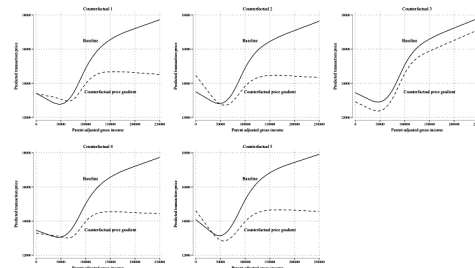
TABLE 10
Comparing baseline and counterfactual pricing patterns at elite colleges

Dependent variable: transaction price	Baseline	Counterfactual				
		1	2	3	4	5
Parent AGI (in \$10,000's)	220.1 (27.86)***	63.8 (25.19)*	44.1 (24.98)	212.0 (26.99)***	60.9 (24.85)*	49.4 (24.69)*
ACT score	-196.6 (47.54)***	-151.9 (48.44)**	-142.6 (48.10)**	-214.2 (49.49)***	-146.5 (49.80)**	-147.8 (48.99)*
High school GPA	-1.087.1 (365.7)*	-1.316.7 (367.5)***	-1.270.5 (366.8)***	-1.160.5 (371.6)**	-1.249.1 (373.3)***	-1.425. (370.7)**
Earned AP credit in high school	-193.8 (404.8)	-19.3 (378.3)	-42.7 (375.8)	-192.3 (376.6)	-93.7 (379.9)	-261.0 (376.0)
Completed FAFSA	-1,110.1 (491.1)*	-1,785.7 (455.3)***	-1,908.2 (454.0)***	-778.9 (458.7)	-939.6 (462.6)*	1,110.6 (455.0)*
Num of add'l colleges listed on FAFSA	-527.9 (112.6)***	-395.6 (114.3)***	-396.1 (113.7)***	-548.8 (112.4)***	-555.9 (114.2)***	-609.7 (115.3)**
Age as of 12/31/07	77.8 (283.8)	112.9 (251.3)	112.4 (251.5)	219.1 (256.8)	126.0 (255.9)	325.4 (256.4)
Female	-553.4 (314.5)	-552.7 (306.8)	-576.3 (307.0)	-629.9 (313.5)*	-578.6 (311.5)	-559.3 (310.9)
Black	-2,168.8 (636.6)***	-3,048.8 (643.6)***	-3,240.4 (639.0)***	-2,710.1 (652.8)***	-3,198.8 (654.1)***	-3,425. (652.7)**
Hispanic	-1,826.3 (638.8)**	-2,351.3 (623.2)***	-2,477.9 (626.7)***	-1,799.0 (618.2)**	-2,084.4 (622.0)***	-2,287. (626.1)**
Asian	-110.6 (822.9)	34.2 (717.0)	3.3 (720.1)	415.9 (708.5)	96.4 (714.9)	-5.8 (711.0)
Other / Multiple	-865.1 (810.3)	-1,503.2 (844.4)	-1,719.1 (828.3)*	-1,358.2 (841.7)	-1,298.6 (820.1)	-1,669. (822.7)*
Out-of-State, Public	8,722.5 (837.8)***	8,804.1 (683.1)***	8,715.0 (676.7)***	8,939.6 (698.0)***	8,576.4 (678.8)***	7,987.8 (628.7)**
College fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,210	2,030	2,020	2,010	1,980	1,930
R ²	0.744	0.772	0.772	0.778	0.778	0.769
FAFSA information available						
Parent income	Yes	Poverty Dummy	No	Yes	Poverty Dummy	No
Number of schools listed on FAFSA	Yes	Yes	Yes	No	No	No
Whether completed FAFSA	Yes	Yes	Yes	Yes	No	No

Notes: All six regressions include students from the elite sample. The dependent variable in column 1 is transaction price (sticker price minus discount). In Columns 2–6, the dependent variable is the counterfactual price the student would have been charged based on the structural estimates. In each counterfactual, students who would be priced out of the elite sector are omitted from the regression. Robust standard errors are in parentheses. Sampling weights were used in all six regressions (NPSAS variable WTA000).

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.



The fitted values plotted here come from five pairs of regressions. In each pair, the fitted values represent a white, female student with all other covariates set to their sample means. Each pair of regressions only include students who were not priced out of the elite college market in the corresponding counterfactual. Parent adjusted gross income was included as a cubic spline.

SOURCE: U.S. Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.

FIGURE 5

(a) Counterfactual pricing patterns

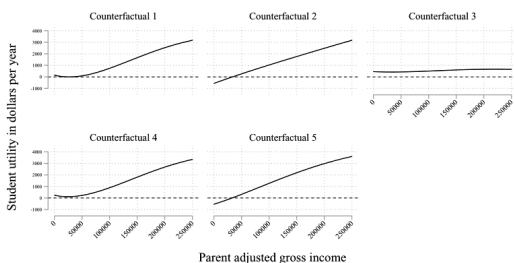
(b) Counterfactual income gradient

5.9 Figure 6 and Tables 11–12: Distributional consequences by income group

Read: High-income students benefit most from restrictions; low-income effects are mixed; only 12.2–22.2% of extracted surplus is transferred to other students.

Interpretation: FAFSA-based price discrimination redistributes on average but imprecisely.

Economic message: The incidence of FAFSA use is heterogeneous within income groups.



The fitted values plotted here come from a separate regression for each counterfactual. In all five regressions, the dependent variable is the estimated change in utility relative to baseline and the independent variable is parent adjusted gross income (included as a fourth-order polynomial). No other covariates are included.

SOURCE: U.S. Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.

FIGURE 6

(a) Change in utility

TABLE 11
Effect of restricting FAFSA info on student welfare and price by income group

		Counterfactual				
		1	2	3	4	5
Bottom third of parent income	Utility change	-\$13	-\$217	\$425	\$115	-\$200
	Price change (if stay at elite college)	\$4	\$221	-\$482	-\$156	\$164
	Percent leaving elite sector	8.8%	10.7%	9.5%	12.2%	16.8%
Middle third of parent income	Utility change	\$262	\$480	\$461	\$397	\$571
	Price change (if stay at elite college)	-\$296	-\$540	-\$529	-\$474	-\$698
	Percent leaving elite sector	8.7%	9.4%	10.6%	12.5%	14.1%
Top third of parent income	Utility change	\$1,615	\$1,755	\$556	\$1,752	\$2,109
	Price change (if stay at elite college)	-\$1,732	-\$1,860	-\$606	-\$1,882	-\$2,28
	Percent leaving elite sector	6.4%	5.4%	6.9%	6.5%	7.1%
Percent of "extracted utility" transferred to other students		18.5%	19.4%	12.2%	14.2%	22.2%
FAFSA information available						
Parent income	Poverty Dum	No	Yes	Poverty Dum	No	
Number of schools listed on FAFSA	Yes	Yes	No	No	No	
Whether completed FAFSA	Yes	Yes	Yes	No	No	

Notes: When colleges can no longer use the FAFSA to price discriminate, some students see their prices fall, relative to baseline, some see their prices rise, and the rest are priced out of the elite market. The three super rows in the table were calculated separately for each of the three income terciles. The first subrow with each super row reports the change in mean utility. The second subrow reports the mean change in transaction price for those students who are not priced out of the market. The third subrow reports the percent of students who are priced out of the elite market. The table also reports the fraction of utility that colleges are able to extract via price discrimination that is transferred to other students. This measures the degree to which colleges use FAFSA information to price discriminate in a way that redistributes for some students to others versus simply increasing college payoffs. No sample weights were used.

Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.

(b) Welfare and price by income group

TABLE 12
Percent of students who benefit from restricting FAFSA info by income group

	Counterfactual (%)				
	1	2	3	4	5
Bottom third of parent income	45.1	33.8	65.9	48.5	33.3
Middle third of parent income	61.9	70.6	72.3	65.8	59.1
Top third of parent income	90.8	91.3	80.3	88.7	86.1
FAFSA information available					
Parent income	Poverty Dum	No	Yes	Poverty Dum	No
Number of schools listed on FAFSA	Yes	Yes	No	No	No
Whether completed FAFSA	Yes	Yes	Yes	No	No

Notes: When colleges can no longer use the FAFSA to price discriminate, some students benefit from lower prices while others are hurt by either higher prices or being priced out of the elite market. Each cell reports the percent of student who benefit, relative to baseline, in each counterfactual in the corresponding tercile of the distribution of parent adjusted gross income. No sample weights were used.
Source: US Department of Education, National Center for Education Statistics, National Postsecondary Student Aid Study, 2008, and Beginning Postsecondary Students, 2004.

is being transferred to other students and how much is accruing to the colleges? The last line o

Percent who benefit from restricting FAFSA info

6 Interpretive synthesis

FAFSA data are valuable because they reveal willingness to pay and perceived competition. Elite colleges use the data to extract surplus, but also to compete and to redistribute imperfectly toward lower-income students. Restricting data access lowers prices for many students but worsens sorting and leaves room for proxy-based price discrimination.

7 Short summary

- The paper studies how FAFSA information affects college price discrimination.
- NPSAS provides student-level prices; BPS measures true applications.
- The structural model is a first-price auction in student utility bids.
- Colleges capture about 69.9% of match surplus.
- Restricting FAFSA information lowers prices for many students but prices some out of elite colleges.
- Average student welfare rises, total surplus falls slightly, and redistribution is imprecise.

8 Conclusion

Fillmore shows that FAFSA information is a powerful pricing technology. Restricting it can raise student welfare through lower prices, but the cost is less efficient matching and only imperfect redistribution. The paper's central message is that need-based information can be used both for aid targeting and for surplus extraction.